FORMULA SHEET FOR STATISTICAL INFERENCE

INFERENCE FOR SINGLE POPULATION

Test Statistics for Mean

Test Statistic $z = \dfrac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ *and* $t = \dfrac{\bar{x}-\mu}{s/\sqrt{n}}$ *with* $d.f = n-1$

Critical Regions and P-Values for Z-test and T-test

	CR for $z-test$	*CR for* $t-test$	$P-values$ *for* $z-test$	$P-values$ *for* $t-test$
If H_1 *contains* $\neq$ *sign*	$\lvert z\rvert \geq z_{\frac{\alpha}{2}}$	$\lvert t\rvert \geq t_{\frac{\alpha}{2}(n-1)}$	$P = 2[1-\Phi(\lvert z_0\rvert)]$	$P = 2[\Phi(\lvert t_0\rvert)]$
If H_1 *contains* $<$ *sign*	$z < -z_{\alpha}$	$t < -t_{\alpha(n-1)}$	$P = \Phi(z_0)$	$P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = \Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$
If H_1 *contains* $>$ *sign*	$z > z_{\alpha}$	$t > t_{\alpha(n-1)}$	$P = 1-\Phi(z_0)$	$P = \Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$

Confidence Intervals for Z-test and T-test

	Confidence Intervals for $z-test$	*Confidence Intervals for* $t-test$
If H_1 *contains* $\neq$ *sign*	$\bar{x} - Z_{\frac{\alpha}{2}}.\dfrac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + Z_{\frac{\alpha}{2}}.\dfrac{\sigma}{\sqrt{n}}$	$\bar{x} - t_{\frac{\alpha}{2}(d.f)}.\dfrac{S}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\frac{\alpha}{2}(d.f)}.\dfrac{S}{\sqrt{n}}$
If H_1 *contains* $<$ *sign*	$\mu \leq \bar{x} + Z_{\alpha}.\dfrac{\sigma}{\sqrt{n}}$	$\mu \leq \bar{x} + t_{\alpha(d.f)}.\dfrac{S}{\sqrt{n}}$
If H_1 *contains* $>$ *sign*	$\mu \geq \bar{x} - Z_{\alpha}.\dfrac{\sigma}{\sqrt{n}}$	$\mu \geq \bar{x} - t_{\alpha(d.f)}.\dfrac{S}{\sqrt{n}}$

INFERENCE FOR TWO POPULATIONS

Test Statistics for Difference of Means

Test Statistic $Z = \dfrac{(\bar{x_1}-\bar{x_2})-(\mu_1-\mu_2)}{\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}}$ (*when* σ_1^2 *and* σ_2^2 *are known*)

$t = \dfrac{(\bar{x_1}-\bar{x_2})-(\mu_1-\mu_2)}{S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}}$ *with* $d.f = n_1+n_2-2$ *where* $S_p = \sqrt{\dfrac{(n_1-1)S_1^2+(n_2-1)S_2^2}{n_1+n_2-2}}$ (*when* σ_1^2 *and* σ_2^2 *are unknown but* $\sigma_1^2 = \sigma_2^2$)

Critical Regions and P-Values

	CR for $z-test$	*CR for* $t-test$	$P-values$ *for* $z-test$	$P-values$ *for* $t-test$
If H_1 *contains* $\neq$ *sign*	$\lvert z\rvert \geq z_{\frac{\alpha}{2}}$	$\lvert t\rvert \geq t_{\frac{\alpha}{2}(n_1+n_2-2)}$	$P = 2[1-\Phi(\lvert z_0\rvert)]$	$P = 2[\Phi(\lvert t_0\rvert)]$
If H_1 *contains* $<$ *sign*	$z < -z_{\alpha}$	$t < -t_{\alpha(n_1+n_2-2)}$	$P = \Phi(z_0)$	$P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = \Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$
If H_1 *contains* $>$ *sign*	$z > z_{\alpha}$	$t > t_{\alpha(n_1+n_2-2)}$	$P = 1-\Phi(z_0)$	$P = \Phi(\lvert t_0\rvert)$ *if* $t_0 > 0$ & $P = 1-\Phi(\lvert t_0\rvert)$ *if* $t_0 < 0$

Confidence Intervals for Difference of Means

	Confidence Intervals for $z-test$
If H_1 *contains* $\neq$ *sign*	$(\bar{x_1}-\bar{x_2}) - Z_{\frac{\alpha}{2}}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}} \leq \mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + Z_{\frac{\alpha}{2}}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
If H_1 *contains* $<$ *sign*	$\mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + Z_{\alpha}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$
If H_1 *contains* $>$ *sign*	$\mu_1-\mu_2 \geq (\bar{x_1}-\bar{x_2}) - Z_{\alpha}.\sqrt{\dfrac{\sigma_1^2}{n_1}+\dfrac{\sigma_2^2}{n_2}}$

	Confidence Intervals for $t-test$
If H_1 *contains* $\neq$ *sign*	$(\bar{x_1}-\bar{x_2}) - t_{\frac{\alpha}{2}(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}} \leq \mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + t_{\frac{\alpha}{2}(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
If H_1 *contains* $<$ *sign*	$\mu_1-\mu_2 \leq (\bar{x_1}-\bar{x_2}) + t_{\alpha(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$
If H_1 *contains* $>$ *sign*	$\mu_1-\mu_2 \geq (\bar{x_1}-\bar{x_2}) - t_{\alpha(n_1+n_2-2)}S_p\sqrt{\dfrac{1}{n_1}+\dfrac{1}{n_2}}$

HAZBER SAMSON

SIMPLE LINEAR REGRESSION AND CORRELATION

$$r=\frac{n\left(\sum xy\right)-\left(\sum x\right)\left(\sum y\right)}{\sqrt{[n\sum x^2-\left(\sum x\right)^2][n\sum y^2-\left(\sum y\right)^2]}} \qquad For\ Testing\ Test\ Statistic\ is\ t=\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}\ \ with\ d.f=n-2$$

$$For\ Y=\alpha+\beta X+\varepsilon,\quad \hat{\beta}=\frac{n\left(\sum xy\right)-\left(\sum x\right)\left(\sum y\right)}{n\left(\sum x^2\right)-\left(\sum x\right)^2}\quad and\quad \hat{\alpha}=\overline{Y}-\hat{\beta}\overline{X}$$

$$R^2=\frac{SSR}{SST}=\frac{\sum\left(\hat{Y}-\overline{Y}\right)^2}{\sum\left(Y-\overline{Y}\right)^2}\quad or\quad R^2=1-\frac{SSE}{SST}=1-\frac{\sum\left(Y-\hat{Y}\right)^2}{\sum\left(Y-\overline{Y}\right)^2}$$

TESTING OF INTERCEPT α

$$Test\ Statistic\ t=\frac{\hat{\alpha}-\alpha}{S.E(\hat{\alpha})}\quad where\ S.E(\hat{\alpha})=\sqrt{\frac{\sigma_e^{\ 2}\sum X^2}{n.\sum\left(X-\overline{X}\right)^2}}\quad also\ \sigma_e^{\ 2}=\frac{\sum e^2}{n-2}\quad CR:\left|t_{cal}\right|\geq t_{\frac{\alpha}{2}(n-2)}$$

The $(1-\alpha)$ 100% for confidence interval for $\hat{\alpha}$ is

$$\hat{\alpha}-t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\alpha})\leq\alpha\leq\ \hat{\alpha}+t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\alpha})$$

TESTING OF INTERCEPT β

$$Test\ Statistic\ \ t=\frac{\hat{\beta}-\beta}{S.E\left(\hat{\beta}\right)}\quad where\ S.E(\hat{\beta})=\sqrt{\frac{\sigma_e^{\ 2}}{\sum\left(X-\overline{X}\right)^2}}\quad also\ \sigma_e^{\ 2}=\frac{\sum e^2}{n-2}\quad and\ CR:\left|t_{cal}\right|\geq t_{\frac{\alpha}{2}(n-2)}$$

The $(1-\alpha)$ 100% for confidence interval for β is

$$\hat{\beta}-t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta})\leq\beta\leq\ \hat{\beta}+t_{\frac{\alpha}{2}(n-2)}.S.E(\hat{\beta})$$

FREQUENTLY USED CRITICAL VALUES OF Z

Significance Level (α)	Confidence Level $(1-\alpha)$	One-Tailed Test	Two-Tailed Test
$\alpha=0.01$	99%	$Z_\alpha=2.33$	$Z_{\alpha/2}=2.58$
$\alpha=0.02$	98%	$Z_\alpha=2.05$	$Z_{\alpha/2}=2.33$
$\alpha=0.03$	97%	$Z_\alpha=1.88$	$Z_{\alpha/2}=2.17$
$\alpha=0.04$	96%	$Z_\alpha=1.75$	$Z_{\alpha/2}=2.05$
$\alpha=0.05$	95%	$Z_\alpha=1.65$	$Z_{\alpha/2}=1.96$
$\alpha=0.06$	94%	$Z_\alpha=1.55$	$Z_{\alpha/2}=1.88$
$\alpha=0.07$	93%	$Z_\alpha=1.48$	$Z_{\alpha/2}=1.81$
$\alpha=0.08$	92%	$Z_\alpha=1.41$	$Z_{\alpha/2}=1.75$
$\alpha=0.09$	91%	$Z_\alpha=1.34$	$Z_{\alpha/2}=1.70$
$\alpha=0.10$	90%	$Z_\alpha=1.28$	$Z_{\alpha/2}=1.65$

HAZBER SAMSON

MULTIPLE LINEAR REGRESSION FORMULAS

For MLRM $Y = XB + E$

1 – *Estimation of Parameters* $\hat{B} = (X^t X)^{-1}(X^t Y)$

2 – *Fitted Values* $\hat{y} = X\hat{\beta}$

3 – *Error Terms* $e = y - \hat{y}$

4 – *Sum of Squares of Errors* $SSE = e^t e$

5 – *Sum of Squares of* Re*gression* $SSR = \hat{B}^t X^t Y - n\overline{Y}^2$

6 – *Sum of Squares Total* $SST = Y^t Y - n\overline{Y}^2$

7 – *Coefficient of Deter*min*ation* $R^2 = \dfrac{SSR}{SST}$ *or* $R^2 = 1 - \dfrac{SSE}{SST}$

8 – *Variance – Co*var*iance Matrix* $C = \sigma_e^2 (X^t X)^{-1}$ *where* $\sigma_e^2 = \dfrac{SSE}{n-p}$

9 – *Correlation Matrix*

$$C.M = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} 1 & r_{12} & r_{13} \\ r_{21} & 1 & r_{23} \\ r_{31} & r_{32} & 1 \end{bmatrix}$$

Note that $r_{11} = r_{22} = r_{33} = 1$ *and* $r_{12} = r_{21}$, $r_{13} = r_{31}$, $r_{23} = r_{32}$

10 – *Testing of Parameters of MLRM*

Test Statistic $t = \dfrac{\hat{\beta}_j - \beta_j}{se(\hat{\beta}_j)}$ *with* $d.f = n - p$

The $(1-\alpha)$ 100% **confidence interval** for $\hat{\beta}_j$ is

$$\hat{\beta}_j - t_{\frac{\alpha}{2}(n-p)} \times se(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{\frac{\alpha}{2}(n-p)} \times se(\hat{\beta}_j)$$

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